A Fractional Product Cover Refinement of Entangled Parallel Repetition in Quantum Games

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August 9, 2026

Abstract

We record a refinement of the exponential parallel-repetition theorem for finite two-player entangled games proved in Chapter 6 of OpenAI's *Ten Advances in Mathematics and Theoretical Computer Science*. The published bound contains the answer-alphabet penalty $\log(|A||B|)$. We show, modulo the Chapter 6 lemmas themselves, that the same proof framework permits replacing this term by $\log r(G)$, where $r(G)$ is the maximum exact fractional product cover number of the acceptance predicate over supported question pairs.

The key step is to augment the repeated-game probability space with an auxiliary product-branch variable

$$Z' \in [r(G)]^D \cup \{\perp\}$$

such that the conditioned winning event is exactly $W_D = \{Z' \neq \perp\}$. On accepted branches the index Z' has at most $r(G)^{|D|}$ values, while the associated weighted local effects remain positive contractions. This preserves the two properties needed by the Chapter 6 quantum martingale and classical history-sampleability arguments without retaining the literal answer transcript (A_D, B_D).

The mathematical argument below is intended to be complete relative to the published Chapter 6 lemmas. Novelty and independent verification remain external questions.

1 Setup

Let

$$G = (X, Y, A, B, \mu, V)$$

be a finite two-player one-round game, with entangled value

$$\omega^*(G) = 1 - \varepsilon < 1.$$

For a supported question pair $(x, y) \in \mathrm{supp}(\mu)$, define the exact fractional product cover number[1] r_{xy} as the least positive integer r for which there exist functions

$$f_{xy,z} : A \to [0,1], \qquad g_{xy,z} : B \to [0,1], \qquad z \in [r],$$

such that

$$V(x,y,a,b) = \sum_{z=1}^{r} f_{xy,z}(a) g_{xy,z}(b) \tag{1}$$

[1] The fractional product cover refinement was arrived at independently from the Chapter 6 argument; the classical analogue appears in Holenstein, Theorem 8.2.

for all $a \in A$ and $b \in B$.

Set

$$r(G) = \max_{(x,y)\in \mathrm{supp}(\mu)} r_{xy}.$$

If necessary, smaller covers are padded with zero functions so that a common product-branch index set $[r(G)]$ may be used.

Remark 1 (Relation to nonnegative rank)**.** *For each supported question pair (x, y), let M_{xy} denote the acceptance matrix*

$$M_{xy}(a, b) = V(x, y, a, b).$$

Since V is Boolean, M_{xy} is a nonnegative $0/1$ matrix. The exact fractional product cover number r_{xy} is equal to the nonnegative rank of this matrix:

$$r_{xy} = \mathrm{rank}_+(M_{xy}).$$

Indeed, both quantities are the minimum number of nonnegative rank-one terms required in an exact decomposition

$$M_{xy}(a, b) = \sum_{z=1}^{r} f_z(a)g_z(b).$$

The fractional product cover definition additionally requires $f_z, g_z \in [0, 1]$. This does not change the minimum for a $0/1$ matrix. In any nonnegative factorization, each individual product satisfies

$$f_z(a)g_z(b) \le M_{xy}(a, b) \le 1.$$

For each nonzero term, rescaling one factor by a positive constant and the other by its reciprocal preserves the product and can be chosen so that both factors take values in $[0, 1]$.

Consequently,

$$r(G) = \max_{(x,y)\in \mathrm{supp}(\mu)} \mathrm{rank}_+(M_{xy}).$$

In particular, since the nonnegative rank of an $|A| \times |B|$ nonnegative matrix is at most $\min\{|A|, |B|\}$,

$$r(G) \le \min\{|A|, |B|\}.$$

Thus the fractional product cover parameter can also be viewed as the maximum nonnegative rank of the game's acceptance matrices.

Remark 2 (Computing the parameter)**.** *To determine $r(G)$, for each supported question pair (x, y), form the acceptance matrix*

$$M_{xy}(a, b) = V(x, y, a, b)$$

and compute its nonnegative rank. Then

$$r(G) = \max_{(x,y)\in \mathrm{supp}(\mu)} \mathrm{rank}_+(M_{xy}).$$

The refinement strengthens the parallel-repetition bound; it is not intended as a more efficient way to compute the game parameter. Computing nonnegative rank can itself be computationally difficult. The gain appears in the exponent: when

$$r(G) \ll |A||B|,$$

the term $\log r(G)$ is smaller than $\log(|A||B|)$, yielding a stronger exponential repetition bound.

2 Candidate theorem

Theorem 1 (Fractional product cover refinement)**.** *There exists a universal constant $c > 0$ such that, for every finite two-player one-round game G as above and every $n \geq 1$,*

$$\omega^*(G^{\otimes n}) \leq \exp\left[-c \, \frac{\varepsilon^{13}}{\varepsilon + \log r(G)} \, n\right] = \exp\left[-\Omega\left(\frac{\varepsilon^{13}}{\varepsilon + \log r(G)} \, n\right)\right].$$

Thus the answer-alphabet term $\log(|A||B|)$ in the published parallel-repetition argument may be replaced by $\log r(G)$.

Remark 3 (Status)**.** *The proof below is relative to the lemmas established in Chapter 6 of the cited OpenAI manuscript. The present note supplies the additional fractional product cover construction and verifies that the history variable and weighted effects satisfy the hypotheses needed by those lemmas. Independent specialist review and a novelty search are still required before the result should be regarded as externally certified or new.*

3 Positive product decomposition of the conditioned winning event

Fix an arbitrary finite-dimensional strategy for $G^{\otimes n}$, with shared state $|\psi\rangle$ and POVMs

$$\{E_{x^n}^{a^n}\}_{a^n \in A^n}, \qquad \{F_{y^n}^{b^n}\}_{b^n \in B^n}.$$

Let $D \subsetneq [n]$ be a conditioning core and let W_D denote the event that all coordinates in D are won. Write

$$m = n - |D|, \qquad p = \mathbb{P}(W_D), \qquad \tau = \log(1/p).$$

For a realized core question tuple (x_D, y_D) and $z_D = (z_j)_{j \in D} \in [r(G)]^D$, define

$$f_{z_D}(a_D) = \prod_{j \in D} f_{x_j y_j, z_j}(a_j), \qquad g_{z_D}(b_D) = \prod_{j \in D} g_{x_j y_j, z_j}(b_j).$$

Define weighted local effects

$$E_{x^n}^{z_D} = \sum_{a^n \in A^n} f_{z_D}(a_D) \, E_{x^n}^{a^n}, \tag{2}$$

$$F_{y^n}^{z_D} = \sum_{b^n \in B^n} g_{z_D}(b_D) \, F_{y^n}^{b^n}. \tag{3}$$

Since all weights lie in $[0, 1]$,

$$0 \preceq E_{x^n}^{z_D} \preceq I, \qquad 0 \preceq F_{y^n}^{z_D} \preceq I.$$

Remark 4. *After conditioning on the revealed questions, each product-cover term gives local positive operators E^{z_D} and F^{z_D}, preserving the tensor-product form required by the Chapter 6 argument. These operators need not themselves form a POVM.*

The key identity is

Lemma 1. *For every question tuple (x^n, y^n),*

$$\sum_{z_D} \langle \psi | E_{x^n}^{z_D} \otimes F_{y^n}^{z_D} | \psi \rangle = \mathbb{P}(W_D \mid x^n, y^n).$$

Proof. By bilinearity,

$$\begin{aligned}
&\sum_{z_D} \langle\psi| E^{z_D}_{x^n} \otimes F^{z_D}_{y^n} |\psi\rangle \\
&= \sum_{a^n,b^n} \mathbb{P}(a^n, b^n \mid x^n, y^n) \prod_{j\in D} \sum_{z_j} f_{x_j y_j, z_j}(a_j) g_{x_j y_j, z_j}(b_j) \\
&= \sum_{a^n,b^n} \mathbb{P}(a^n, b^n \mid x^n, y^n) \prod_{j\in D} V(x_j, y_j, a_j, b_j),
\end{aligned}$$

and the last expression is exactly the conditional probability of W_D. □

Thus the accepted core event is decomposed into at most

$$r(G)^{|D|}$$

positive product branches.

4 Replacement of the answer transcript

The published proof carries the literal core-answer word

$$(A_D, B_D) \in A^D \times B^D,$$

whose logarithmic size is

$$|D| \log(|A||B|).$$

The product decomposition above permits replacing this word by the branch label

$$Z_D \in [r(G)]^D,$$

whose logarithmic size is

$$s_r := |D| \log r(G).$$

It remains to verify that the product witness can replace the literal answer transcript throughout the proof.

5 Ideal branch and local POVM refinement

Let T denote the same revealed-question history used in the published proof, and let $i \notin D$ be the live coordinate. For a history $r = (t, z_D)$, define conditional averaged effects

$$H_{r,x} = \mathbb{E}\big[E^{z_D}_{X^n} \mid T = t, X_i = x\big], \qquad K_{r,y} = \mathbb{E}\big[F^{z_D}_{Y^n} \mid T = t, Y_i = y\big].$$

Define

$$p_r(x, y) = \langle\psi| H_{r,x} \otimes K_{r,y} |\psi\rangle.$$

The corresponding posterior branch distribution is

$$Q'(i, r, x, y) = \frac{1}{p} P^0(i, t, x, y)\, p_r(x, y),$$

where P^0 is the prior distribution over the revealed-question history and live questions appearing in the published proof.

To recover the live answer distribution, refine $E^{z_D}_{x^n}$ according to the live answer:

$$E^{z_D,a}_{x^n} = \sum_{\substack{a^n \\ a_i=a}} f_{z_D}(a_D) E^{a^n}_{x^n},$$

and analogously on Bob's side. Averaging gives positive refinements

$$H^a_{r,x} \succeq 0, \qquad \sum_a H^a_{r,x} = H_{r,x},$$

and similarly

$$K^b_{r,y} \succeq 0, \qquad \sum_b K^b_{r,y} = K_{r,y}.$$

Therefore the singular-safe local POVM refinement lemma from Chapter 6 applies without change. The ideal branch reproduces the live-coordinate answer distribution associated with the weighted branch. Summing over z_D recovers the original live-answer distribution conditioned on W_D. Hence the average ideal live-coordinate winning probability is the same quantity

$$q = \frac{1}{m} \sum_{i \notin D} \mathbb{P}(W_i \mid W_D)$$

used in the published proof.

6 Auxiliary product-branch variable

The published proof retains the literal core answer word (A_D, B_D). Here we replace it with an auxiliary variable

$$Z' \in [r(G)]^D \cup \{\perp\}.$$

Conditioned on a complete core realization (x_D, y_D, a_D, b_D), define for $z_D \in [r(G)]^D$

$$\mathbb{P}\big(Z' = z_D \mid x_D, y_D, a_D, b_D\big) = \prod_{j \in D} f_{x_j y_j, z_j}(a_j) g_{x_j y_j, z_j}(b_j), \tag{4}$$

and assign the remaining probability to the failure symbol $\perp$.

Because the acceptance predicate is Boolean and the factors in the exact fractional product cover are nonnegative,

$$\sum_{z_j} f_{x_j y_j, z_j}(a_j) g_{x_j y_j, z_j}(b_j) = V(x_j, y_j, a_j, b_j) \in \{0, 1\}.$$

Hence

$$\sum_{z_D \in [r(G)]^D} \mathbb{P}(Z' = z_D \mid x_D, y_D, a_D, b_D) = \prod_{j \in D} V(x_j, y_j, a_j, b_j).$$

Therefore

$$\boxed{W_D = \{Z' \neq \perp\}} \tag{5}$$

as events on the augmented probability space.

Thus the accepted branch variable determines W_D exactly, as does the literal answer transcript in the published proof. Moreover, for each accepted branch z_D its probability equals the weighted positive product mass

$$\langle\psi|E^{z_D}_{x^n} \otimes F^{z_D}_{y^n}|\psi\rangle,$$

by the bilinear calculation already proved above.

Lemma 2 (Quantum-transfer verification)**.** *The weighted effects $E^{z_D}_{X^n}$ and $F^{z_D}_{Y^n}$ satisfy the hypotheses of the Chapter 6 positive-contraction martingale/resolvent argument.*

Proof. Each weighted effect is a positive contraction. Successive conditional expectations along the reveal filtration therefore form positive-contraction Doob martingales by linearity of conditional expectation. Equation (5) and the branch-mass identity show that the accepted product-branch masses sum exactly to the probability of W_D. Thus the entropy telescope and random-live-coordinate argument may be applied branchwise exactly as in Chapter 6. □

Lemma 3 (Classical-history transfer verification)**.** *The Chapter 6 history-sampleability argument remains valid with the accepted product-branch index $Z_D := Z'|_{W_D}$ in place of the literal answer word.*

Proof. Under the posterior $Q' = P(\,\cdot \mid W_D)$, the failure symbol $\perp$ has zero mass and Z_D takes at most $r(G)^{|D|}$ values. The marginal distribution of all referee questions is the same posterior question law obtained by conditioning on W_D, so data processing gives the same conditioning cost $\tau = \log(1/p)$ as in the published proof. Comparing Z_D with the uniform reference distribution on $[r(G)]^D$ adds at most $|D|\log r(G)$ to the relative entropy. The reference measure leaves the original referee question prior unchanged; hence the unrevealed questions retain the same conditional-independence assumptions used in the reveal-chain argument. The subsequent chain-rule, Pinsker, and finite correlated-sampling steps depend only on these joint distributions and support bounds, not on whether Z_D is a literal measurement transcript. □

7 Quantum entropy budget

For each fixed product branch z_D, successive conditional expectations of the positive contractions $E^{z_D}_{X^n}$ and $F^{z_D}_{Y^n}$ form the same kind of positive-contraction Doob martingales used in Chapter 6. Thus the resolvent-purification martingale lemma applies branchwise without change.

Let $p_z(U)$ denote the unnormalized branch mass for background U. Then

$$p_z(U) \geq 0, \qquad \sum_{z_D} p_z(U) = p_W(U),$$

and the number of product-branch indices is at most $r(G)^{|D|}$. Therefore the same log-sum argument as in the accepted-word entropy lemma gives

$$\sum_{z_D} H_1(p_z(U)) \leq H_1(p_W(U)) + p_W(U)\,|D|\log r(G),$$

where

$$H_1(t) = -t\log t.$$

Averaging,

$$\mathbb{E}_U \sum_{z_D} H_1(p_z(U)) \leq p\big(\tau + s_r\big).$$

Consequently the same size-biased random-reveal argument yields

$$I_A, I_B \le \frac{2p(\tau + s_r)}{m}.$$

Define

$$\eta_r = \frac{\tau + s_r}{m} = \frac{\tau + |D| \log r(G)}{m}.$$

The normalization argument of Chapter 6 cancels the explicit factor p and gives

$$\mathbb{E}_{Q'} \|\Psi - \Psi^A\|^2 \le 8\eta_r, \qquad \mathbb{E}_{Q'} \|\Psi - \Psi^B\|^2 \le 8\eta_r.$$

8 Classical history sampleability

Define

$$J'_A(i, r, x, y) = \frac{1}{m} \mu(x, y)\, Q'(r \mid i, X_i = x),$$

and similarly J'_B.

The marginal of Q' on the question variables is the same conditioned question law arising from W_D. Hence the same data-processing argument gives

$$D(Q'_{\text{questions}} \| P_{\text{questions}}) \le \tau.$$

Conditionally on the questions, the witness Z_D has at most $r(G)^{|D|}$ values. Comparing it with the uniform distribution on $[r(G)]^D$ gives

$$D\big(Q'_{U,Y_C,Z_D} \big\| P_{U,Y_C} \otimes \mathrm{Unif}([r(G)]^D)\big) \le \tau + s_r.$$

The reference distribution retains the original question prior, so the unrevealed questions preserve the same conditional-independence relations used in the reveal-chain proof of the published history lemma. Repeating that argument gives

$$D(Q' \| J'_A),\ D(Q' \| J'_B) \le \frac{3\tau + 2s_r}{m}.$$

Pinsker therefore yields

$$d_{\mathrm{TV}}(Q', J'_A),\ d_{\mathrm{TV}}(Q', J'_B) \le \sqrt{\frac{3}{2}\, \eta_r}.$$

The same finite classical correlated-sampling lemma then supplies approximately matching histories.

9 Rounding

All quantitative hypotheses used by the published single-game rounding argument now hold with η replaced by η_r: the ideal branch wins with average probability q; the two locally describable candidate states are close in expected squared distance; and the locally sampled histories are close in total variation.

Therefore the same rounding proof yields

$$\omega^*(G) \ge q - B_{\mathrm{qs}} \eta_r^{1/12},$$

after taking the finite-catalyst approximation parameter to zero.

10 Final contradiction

Let

$$\ell_r = \log r(G), \qquad \gamma = c_{\mathrm{qs}} \frac{\varepsilon^{13}}{\varepsilon + \ell_r}, \qquad \delta = \frac{\varepsilon}{4}.$$

Assume for contradiction that

$$\omega^*(G^{\otimes n}) > e^{-\gamma n}.$$

Choose a repeated strategy with all-coordinate success

$$\vartheta > e^{-\gamma n}.$$

The greedy conditioning lemma used in Chapter 6 gives a proper $D \subsetneq [n]$ with

$$|D| < \frac{\gamma n}{\delta}, \qquad m > \frac{n}{2}, \qquad p \geq \vartheta, \qquad q \geq 1 - \frac{\varepsilon}{4}.$$

Since $\tau = \log(1/p) < \gamma n$,

$$\eta_r < 2\gamma \left(1 + \frac{4\ell_r}{\varepsilon}\right).$$

Choosing the universal implicit constant in the exponent sufficiently small (as in the final parameter calculation of Chapter 6), one obtains

$$\eta_r < \left(\frac{\varepsilon}{4B_{\mathrm{qs}}}\right)^{12}.$$

Hence

$$\omega^*(G) \geq q - B_{\mathrm{qs}} \eta_r^{1/12} > 1 - \frac{\varepsilon}{4} - \frac{\varepsilon}{4} > 1 - \varepsilon,$$

contradicting

$$\omega^*(G) = 1 - \varepsilon.$$

This proves Theorem 1 relative to the Chapter 6 lemmas invoked above. □

11 Interpretation

The published answer-alphabet factor

$$\log(|A||B|)$$

arises because the proof stores the literal conditioned answer word. The argument above shows that the proof can replace the literal conditioned answer transcript by an auxiliary positive product-branch index for acceptance. Thus the bound need not depend on the total number of possible measurement outcomes. In this proof, that dependence can instead be expressed through $r(G)$, the maximum nonnegative rank of the verifier's acceptance matrices.

The parameter $r(G)$ is sufficient for this proof, but is not claimed to be necessary; stronger bounds may be available for particular classes of games.

12 Status and open checks

The auxiliary variable Z' is defined so that

$$W_D = \{Z' \neq \perp\}.$$

Thus $Z' \neq \perp$ exactly when all coordinates in D are won. The remaining open checks are external:

1. Independent specialist verification of the argument, especially the use of the Chapter 6 martingale/resolvent and history-sampleability lemmas with the weighted product-branch family.
2. To the best of our knowledge, the classical analogue appears in Holenstein, Theorem 8.2, and we are not aware of a prior quantum analogue of the proposed refinement.
3. Verification of bibliographic and version details for the cited Chapter 6 manuscript and Holenstein's exact conventions.
4. Only after the clean refinement is settled: optional investigation of distribution-sensitive or other quantitative improvements.

A Example: the CHSH_q game

Let q be a prime power and consider the game CHSH_q. Alice and Bob receive $x, y \in \mathbb{F}_q$ and return $a, b \in \mathbb{F}_q$. The verifier accepts precisely when

$$a + b = xy.$$

For fixed x, y, the acceptance matrix is

$$M_{xy}(a, b) = \mathbf{1}[a + b = xy].$$

For every $a \in \mathbb{F}_q$, there is exactly one accepted value of b, namely

$$b = xy - a.$$

Hence M_{xy} is a $q \times q$ permutation matrix.
It has the explicit nonnegative rank-one decomposition

$$M_{xy} = \sum_{a \in \mathbb{F}_q} e_a e_{xy-a}^{\mathsf{T}},$$

where e_a denotes the standard basis vector indexed by a. Therefore

$$\mathrm{rank}_+(M_{xy}) \leq q.$$

Conversely, every $q \times q$ permutation matrix has ordinary matrix rank q, while nonnegative rank is always at least ordinary rank. Thus

$$\mathrm{rank}_+(M_{xy}) = q.$$

This holds for every question pair (x, y), so

$$r(\mathrm{CHSH}_q) = q.$$

Since both answer alphabets have size q,

$$|A||B| = q^2.$$

The answer-alphabet term in the published parallel-repetition bound is therefore

$$\log(|A||B|) = \log(q^2) = 2\log q,$$

whereas the fractional-product-cover refinement gives

$$\log r(G) = \log q.$$

Consequently, for this family the proposed theorem replaces

$$\frac{\varepsilon^{13}}{\varepsilon + 2\log q}$$

by

$$\frac{\varepsilon^{13}}{\varepsilon + \log q}.$$

For large q relative to the gap term ε, this amounts asymptotically to a factor of approximately 2 improvement in this part of the exponent.

Computational remark. This example also illustrates that determining $r(G)$ need not require materializing the full q^2-entry acceptance matrix. The succinct verifier predicate

$$a + b = xy$$

immediately implies that each fixed-(x, y) acceptance matrix is a permutation matrix, from which $r(G) = q$ follows directly. Thus the refinement does not provide a general efficient algorithm for computing $r(G)$, but for games whose verifier has a succinct description, $r(G)$ can sometimes be determined or bounded without enumerating every answer pair.

References

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